

$$\epsilon_{ijk} = \begin{cases} 1 & \text{جائزہ 123} \\ -1 & \text{123 کا پرت} \\ 0 & \text{دوسرے صورت} \end{cases}$$

$$i, j, k = 1, 2, 3$$

$$123 = \sigma_{123}$$

$$132 = \sigma_{132}$$

$$\delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases} \quad \vec{\delta}_{ij} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sum_k \epsilon_{ijk} \epsilon^{k\ell m} = \delta^{\ell}_i \delta^m_j - \delta^{\ell}_j \delta^m_i$$

$$k=1 \Rightarrow \frac{1}{4\pi\epsilon_0} \rightarrow \epsilon_0 = \frac{1}{4\pi} \quad \mu_0 \epsilon_0 = \frac{1}{c^2} \rightarrow \mu_0 = \frac{4\pi}{c^2} \quad E \rightarrow E \quad B \rightarrow \frac{B}{c}$$

$$\checkmark \left\{ \begin{array}{ll} \nabla \cdot E = 4\pi\rho & \nabla \cdot B = 0 \\ \nabla \times B = \frac{4\pi}{c} \vec{J} + \frac{1}{c} \frac{\partial E}{\partial t} & \nabla \times E = -\frac{1}{c} \frac{\partial B}{\partial t} \end{array} \right\} \checkmark$$

$$\nabla \cdot B = 0 \Rightarrow B = \nabla \times A \quad \nabla \times E = -\frac{1}{c} \frac{\partial}{\partial t} \nabla \times A \rightarrow \nabla \times \left(E + \frac{1}{c} \frac{\partial A}{\partial t} \right) = 0 \Rightarrow$$

$$E + \frac{1}{c} \frac{\partial A}{\partial t} = -\nabla \phi$$

$$x^\mu = (x^0, x^1) = (ct, x, y, z) \quad \partial_\mu = \frac{\partial}{\partial x^\mu} = (\partial_0, \partial_i) \quad \partial_0 = \frac{\partial}{\partial ct}$$

$$(\nabla \times C)^i = \epsilon^{ijk} \partial_j C_k \quad (A \times B)^i = \epsilon^{ijk} A_j B_k \quad \nabla = \partial_i \quad j^\mu = (c\rho, \vec{J})$$

$$\nabla \cdot C = \partial_i C^i = \partial_j C^i \frac{\partial^j}{\partial x^i}$$

$$\partial_i E^i = \frac{4\pi}{c} J^0$$

$$\partial_i B^i$$

$$\epsilon^{ijk} \partial_j B_k = \frac{4\pi}{c} J^i + \partial_0 E^i$$

$$\epsilon^{ijk} \partial_j E_k = -\partial_0 B^i$$

$$F^i = q (\epsilon^{ijk} v_j B_k + E^i) = \int \rho (\epsilon^{ijk} v_j B_k + E^i) dV \Rightarrow$$

$$F^i = \rho (\epsilon^{ijk} v_j B_k + E^i)$$